

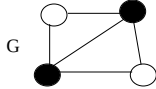
Reduction of 3-SAT to Vertex Cover (VC)

Let G be an undirected graph. A **vertex cover** of G is a subset $COVER$ of V such that for every $(u, v) \in E$, at least one of u or $v \in COVER$.

VC: Vertex Cover Problem

instance: a graph $G=(V, E)$ and a positive integer $k \leq V$

question: is there a subset $COVER \subseteq V$ of size k such that each edge in E has at least one endpoint in $COVER$?



$k = 2$ ($COVER$ = the black nodes) is a "yes" instance for this G .

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Reduction of 3-SAT to VC

The 3-SAT problem is one of the most common NPC problems used on the left side of polynomial time reductions.

The transformation involves taking a boolean formula that would be a "yes" instance to 3-SAT and converting each clause to a set of nodes and edges that are used as an instance of the VC problem.

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Showing VC is NPC

1. Show VC $\in$ NP.

Given an instance and a certificate, the validation requires checking the ends of each edge to see if one end is in $COVER$. In an n node graph, there are $O(n^2)$ edges, so this checking can be done in p -time in the number of edges.

2. Show VC is NP-hard. (show $3\text{-SAT} \leq_p VC$)

Given an instance C of a 3 CNF formula (clauses and variables), construct a graph G and positive integer k such that G has a set $COVER$ of size k iff C is satisfiable.

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Longest Simple Cycle problem

LSC: Longest simple cycle

instance: a graph $G=(V, E)$ and a positive integer $k < V$

question: Is there a simple cycle in G such that the length of the cycle is equal to k ?

Show LSC is NP-complete:

1. Show LSC $\in$ NP.

Given a description of G and a candidate cycle, we can check the number of nodes in the cycle is k , no vertex is repeated (except first and last), and an edge exists between each consecutive pair of vertices in time polynomial in the size of the cycle.

2. Show LSC is NP-hard. (show $HC \leq_p LSC$)

Given an instance G of HC , we transform G into an instance G', k of LSC .

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Showing Clique is NPC

Clique: Clique Problem

instance: a graph $G=(V, E)$ and a positive integer $k \leq V$

question: is there a subset $CLIQUE \subseteq V$ of size k such that $CLIQUE$ is a complete graph?

1. Show Clique $\in$ NP.

Given G and a candidate subset, it is easy to check in p -time that there are k vertices in the subset, and that there are edges between each pair. This takes $O(k^2)$ time.

2. Show Clique is NP-hard. (show $SAT \leq_p \text{Clique}$)

Given an instance C of a CNF formula (clauses and variables), construct a graph G and positive integer k such that G has a set $CLIQUE$ of size k iff C is satisfiable.

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Using related algorithms to prove p -time solutions

LPL: Longest Path Length optimization problem

instance: a graph $G=(V, E)$ and 2 nodes u and v .

question: what is length of longest simple path from u to v ?

LP: Longest Path decision problem

instance: a graph $G=(V, E)$, 2 nodes u and v , and a positive integer, k .

question: is there a simple path from u to v in G whose length is at least k ?

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Using related algorithms to prove p-time solutions

Show that the LPL problem can be solved in p-time iff $LP = P$

(a) If $LP = P$, then we can construct a p-time algorithm for LPL as follows:

- 1) input (G, u, v)
- 2) for $k = n-1$ down to 0 do
 run LP algorithm on (G, u, v, k)
 if result is "yes", return k

(b) If $LPL = P$, then we can construct a p-time algorithm for LP as follows:

- 1) input (G, u, v, k)
- 2) $x =$ result of running LPL algorithm on (G, u, v)
- 3) if $x \geq k$ return "yes"
- 4) else return "no"

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